

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE Regular Summer Examination – 2023 Course: B. Tech. Branch :Civil Semester :VI Subject Code & Name: BTCVC601 Design of RC Structures (DRCS) Max Marks: 60 Date:12/07/2023 Duration: 3 Hr.			
Instructions to the Students: 1. All the questions are compulsory. 2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question. 3. Use of non-programmable scientific calculators is allowed. 4. Assume suitable data wherever necessary and mention it clearly.			
		(Level/CO)	Marks
Q. 1	Solve Any Two of the following.		12
A)	Differentiate between Working stress method and Limit State Method	CO-1	6
B)	Explain different types of loads acting on RC Structure with there IS code Provision.	CO-4	6
C)	Explain Stress strain behavior of Steel & Concrete.	CO-1	6
Q.2	Solve Any Two of the following.		12
A)	Explain the Concept of Under Reinforced, Balanced & Over Reinforced Section.	CO-1	6
B)	An RC Beam 300mm wide & 700 mm deep overall is reinforced with 20mm-3 bars. The centers of bars are 50 mm from underside of beam. The maximum stresses are not to exceed 7 N/mm ² for Concrete and 190 N/mm ² for steel. By using Working stress method, Find Moment of Resistance and Safely distributed load the beam can carry. Span is 6m & Weight of concrete is 25 KN/m ³ .Use Working State Method.(WSM)	CO-3	6
C)	Design the Reinforcement for RC column of a 350 mm x 450 mm in cross section to carry axial load 1000KN.Assume grade of concrete M-20 and Steel Fe415. Permissible stresses in Steel and Concrete are 190 N/mm ² & 5 N/mm ² .Use Working Stress Method.(WSM)	CO-2	6
Q. 3	Solve Any Two of the following.		12
A)	Explain the term – a) Load Factor b) Limit State c) Partial Safety Factor	CO-1	6

B)	Design Shear Reinforcement for beam of size 250 mm x 440 mm overall with cover 40 mm of span 4.25m carries udl 17.5 KN/m including self-weight. Beam is reinforced with 3-16 mm diameter bar. Use M20 Concrete and Fe 415 grade steel. Use Limit State Method.(LSM)	CO-2	6
C)	Find Lap length for bars of 18mm diameter in tension as well compression. Use M20 Concrete & Fe500 grade Steel, Take $\tau_{bd} = 1.2 \text{ N/mm}^2$. Use Limit State Method.	CO-2	6
Q.4	Solve Any Two of the following.		12
A)	A Hall has dimensions 3m x 9m with wall thickness 230mm. The live load on slab is 3 KN/m ² & finishing load of 1 KN/m ² may be assumed. Using M20 grade concrete & Fe 415 grade Steel, Design Slab. Use Limit State Method.	CO-2	6
B)	A T-Beam has permissible effective flange width 1500 mm. Thickness of floor slab 120mm & width of rib is 300mm, effective depth of section 500 mm is reinforced with 4-25mm ϕ . Calculate factored moment of Resistance. Use M25 Concrete & Fe 415 Grade Steel. Use Limit State Method.	CO-3	6
C)	State and Explain different parts of Staircase with neat sketch.	CO-3	6
Q. 5	Solve Any Two of the following.		12
A)	Design a Column , 4m long restrained in position & direction at both ends to carry an axial load of 1600 KN. Use M20 Concrete & Fe415 grade Steel. Use Limit State Method.	CO-2	6
B)	Explain with neat sketch importance of Interaction diagram for uniaxial bending & Axially Compression member.	CO-3	6
C)	Define Footing. Explain different types of Footing.	CO-2	6
	*** End ***		